Japan's Spacecraft Heads Off to Grab Pieces of a Mars Moon

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Japan's **space agency**, JAXA, planned to launch a **mission** to Phobos on Monday. Phobos is one of the two moons of Mars. People could watch the launch live online.

The **mission** is called MMX. Its job is to collect bits of Phobos and bring them back to Earth. Scientists want to learn how the moons of Mars were made. Some think they were once **asteroids**. Others think they formed from leftover space material long ago.

An American **astronomer** named Asaph Hall found Phobos and the smaller moon, Deimos, in 1877. Phobos means "fear" in ancient Greek, and Deimos means "dread."

NASA helped by giving **instruments** and technology.

The spacecraft was set to blast off on an H3 rocket from Japan at 3:41 p.m. Eastern time on October 19. It is a long trip. MMX should reach Phobos in August 2027. The **samples** will not come back to Earth until at least 2031.

Word Treasure

space agency -- A group in a country's government that explores space.
mission -- A special trip with an important job to do.
asteroids -- Space rocks that travel around the Sun.
astronomer -- A scientist who studies stars, planets and moons.
instruments -- Tools that measure or study things, like cameras on a spacecraft.
samples -- Small pieces collected so scientists can study them.

Background you need

Mars is a red planet with two small moons, Phobos and Deimos.

Some spacecraft go to other worlds, scoop up bits of rock and bring them home.

Trips to Mars take a long time because it is very far from Earth.

Break it down

WHO: Japan's space agency, JAXA, with help from NASA
WHAT: A spacecraft called MMX will collect bits of a Mars moon
WHERE: Phobos, one of the two moons of Mars
WHY: To learn how the moons of Mars were made

Why it matters

Space rocks brought home can help us learn how our neighborhood in space formed, and today's students could study them one day.

Test what you remember

1. Where is the MMX mission going?

- (A) Earth's Moon
- (B) The Sun
- (C) Phobos, a moon of Mars
- (D) Jupiter

2. What is the mission's job?

- (A) To collect bits of Phobos and bring them to Earth
- (B) To take people to Mars
- (C) To build a hotel
- (D) To find water on Venus

3. What does Phobos mean in ancient Greek?

- (A) Fear
- (B) Moon
- (C) Dread
- (D) Red

4. Who found Phobos and Deimos in 1877?

- (A) Carson Bruns
- (B) Katie Boulter
- (C) Lord Kelvin
- (D) Asaph Hall

5. How did NASA help?

- (A) It named the moon
- (B) It gave instruments and technology
- (C) It drove the rocket
- (D) It did nothing

6. When will the samples come back to Earth, at the earliest?

- (A) 2031
- (B) 2026
- (C) 1877
- (D) Next week

Answer key (try the quiz first!): 1: C 2: A 3: A 4: D 5: B 6: A

Questions to think about

MY THOUGHTS (at least 20 words):